CHEM 13 · MOLECULE CARD

01

Propane

IUPAC: propane

1 MOLECULAR FORMULA

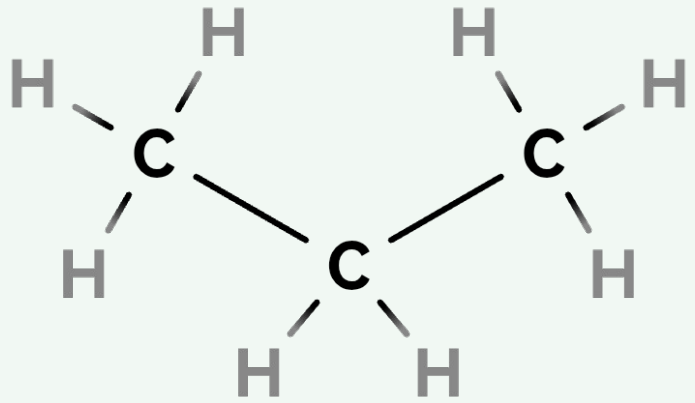
C_3H_8

ATOM COUNT

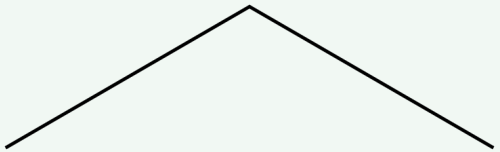
C 3 | H 8

11 atoms

2 LEWIS STRUCTURE



4 BOND-LINE (KETCHER)



3 CONDENSED STRUCTURAL FORMULA

$CH_3CH_2CH_3$

- ✓ atom count verified
- ✓ valence checked
- ✓ bond order checked
- ✓ same connectivity confirmed

ChE connection: LPG and fuel processing

CHEM 13 · MOLECULE CARD

02

Ethanol

IUPAC: ethanol

1 MOLECULAR FORMULA

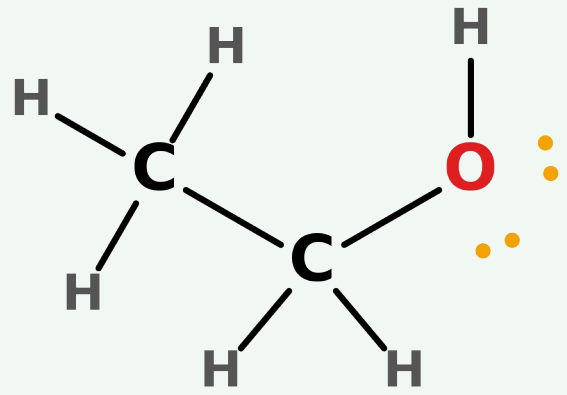
C_2H_6O

ATOM COUNT

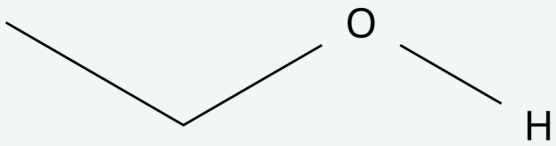
C 2 | H 6 | O 1

9 atoms

2 LEWIS STRUCTURE



4 BOND-LINE (KETCHER)



3 CONDENSED STRUCTURAL FORMULA

CH_3CH_2OH

- ✓ atom count verified
- ✓ valence checked
- ✓ bond order checked
- ✓ same connectivity confirmed

ChE connection: Fermentation, distillation, solvents, biofuels

CHEM 13 · MOLECULE CARD

03

Acetone

IUPAC: propan-2-one

1 MOLECULAR FORMULA

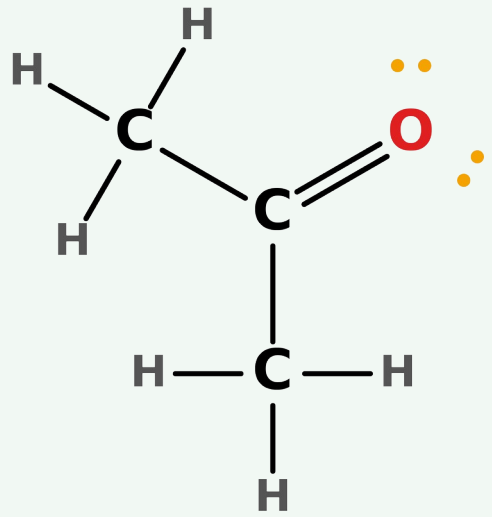
C_3H_6O

ATOM COUNT

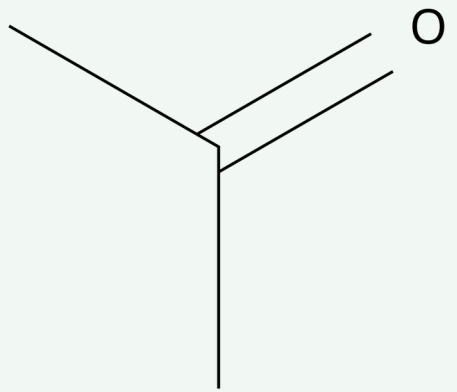
C 3 | H 6 | O 1

10 atoms

2 LEWIS STRUCTURE



4 BOND-LINE (KETCHER)



3 CONDENSED STRUCTURAL FORMULA

$CH_3C(=O)CH_3$

Also written CH_3COCH_3

- ✓ atom count verified
- ✓ valence checked
- ✓ bond order checked
- ✓ same connectivity confirmed

ChE connection: Industrial solvent and solvent recovery

CHEM 13 · MOLECULE CARD

04

Acetic acid

Systematic name: ethanoic acid

1 MOLECULAR FORMULA

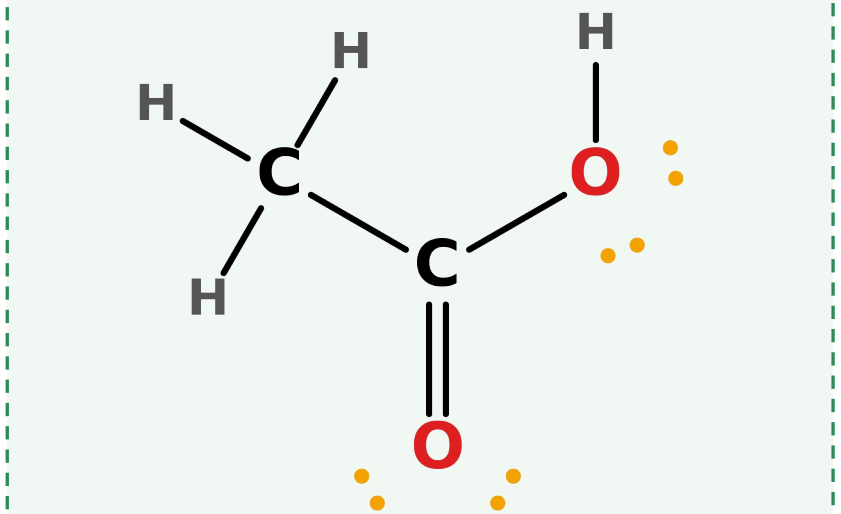
$C_2H_4O_2$

ATOM COUNT

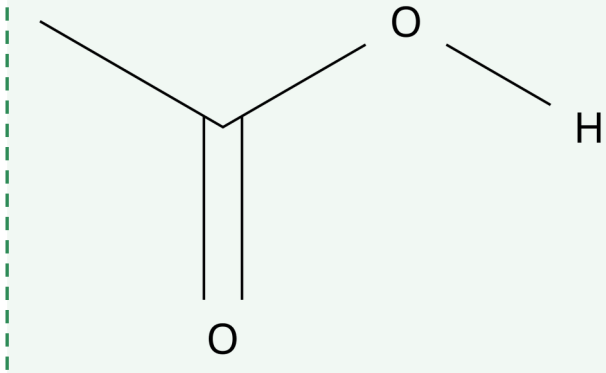
C 2 | H 4 | O 2

8 atoms

2 LEWIS STRUCTURE



4 BOND-LINE (KETCHER)



3 CONDENSED STRUCTURAL FORMULA

$CH_3C(=O)OH$

Also written CH_3COOH

- ✓ atom count verified
- ✓ valence checked
- ✓ bond order checked
- ✓ same connectivity confirmed

ChE connection: Chemical manufacture and acid processing

CHEM 13 · MOLECULE CARD

05

Ethyl acetate

Systematic name: ethyl ethanoate

1 MOLECULAR FORMULA

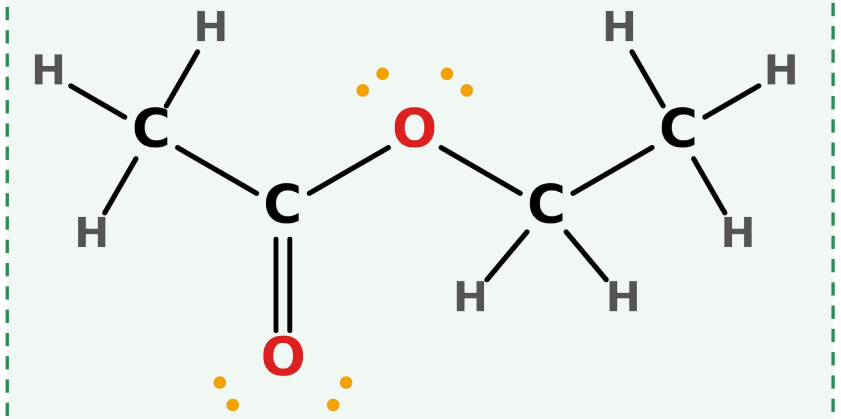
$C_4H_8O_2$

ATOM COUNT

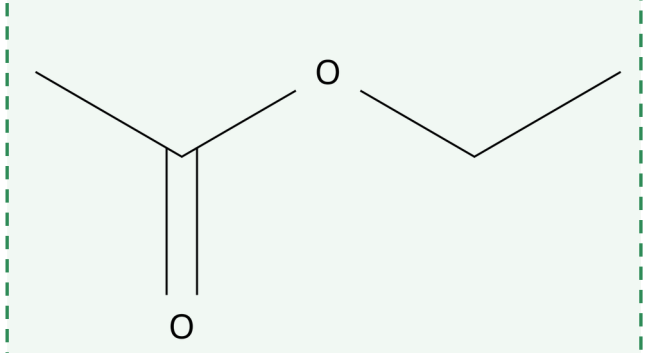
C 4 | H 8 | O 2

14 atoms

2 LEWIS STRUCTURE



4 BOND-LINE (KETCHER)



3 CONDENSED STRUCTURAL FORMULA

$CH_3C(=O)OCH_2CH_3$

Acetyl group joined through O to the ethyl group

- ✓ atom count verified
- ✓ valence checked
- ✓ bond order checked
- ✓ same connectivity confirmed

ChE connection: Esterification, solvents, extraction

CHEM 13 · MOLECULE CARD

06

Cyclohexane

IUPAC: cyclohexane

1 MOLECULAR FORMULA

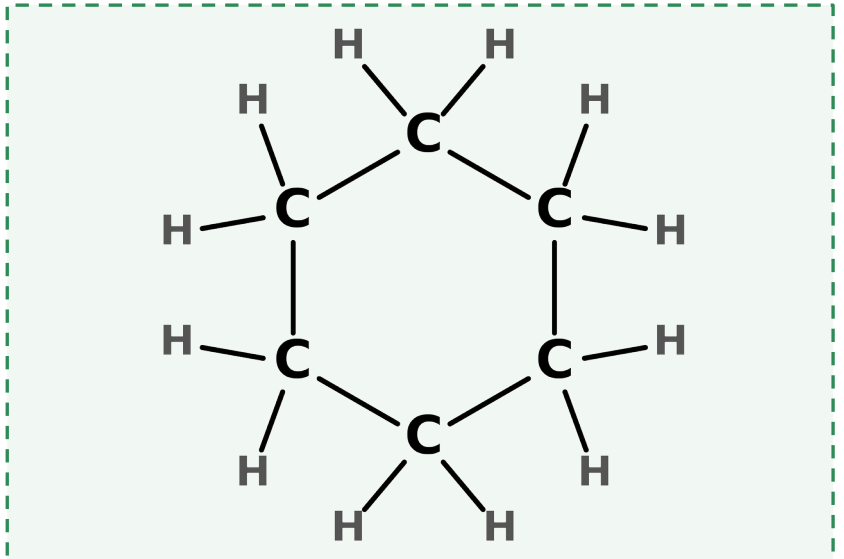
C_6H_{12}

ATOM COUNT

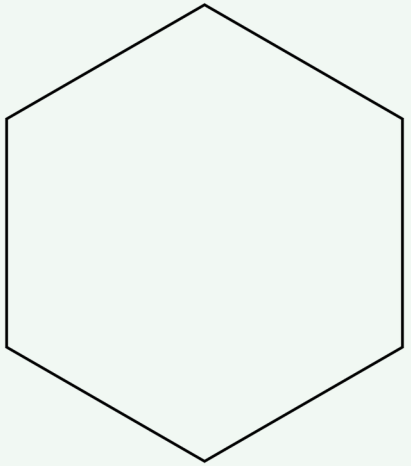
C 6 | H 12

18 atoms

2 LEWIS STRUCTURE



4 BOND-LINE (KETCHER)



3 CONDENSED STRUCTURAL FORMULA

$$\text{cyclo-}(CH_2)_6$$

Cyclic: six CH_2 units joined in a ring

- ✓ atom count verified
- ✓ valence checked
- ✓ bond order checked
- ✓ same connectivity confirmed

ChE connection: Petrochemical intermediate

CHEM 13 · MOLECULE CARD

07

Cyclohexanone

IUPAC: cyclohexanone

1 MOLECULAR FORMULA

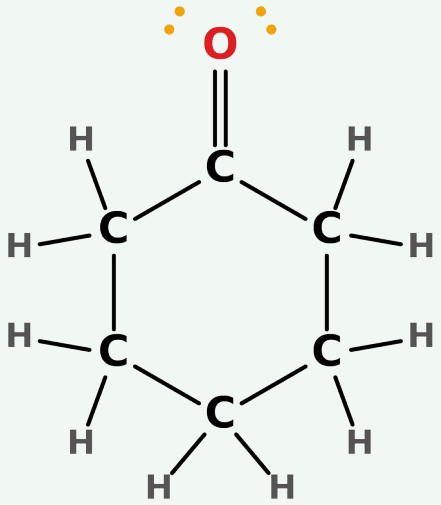
$C_6H_{10}O$

ATOM COUNT

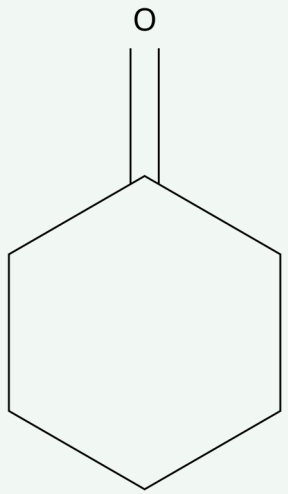
C 6 | H 10 | O 1

17 atoms

2 LEWIS STRUCTURE



4 BOND-LINE (KETCHER)



3 CONDENSED STRUCTURAL FORMULA

cyclo-C(=O)(CH$_2$)$_5$

Cyclic: one C=O carbon and five CH_2 units in a ring

- ✓ atom count verified
- ✓ valence checked
- ✓ bond order checked
- ✓ same connectivity confirmed

ChE connection: Intermediate in nylon-related manufacturing

CHEM 13 · MOLECULE CARD

08

Acetaminophen (paracetamol)

IUPAC: N-(4-hydroxyphenyl)acetamide

1 MOLECULAR FORMULA

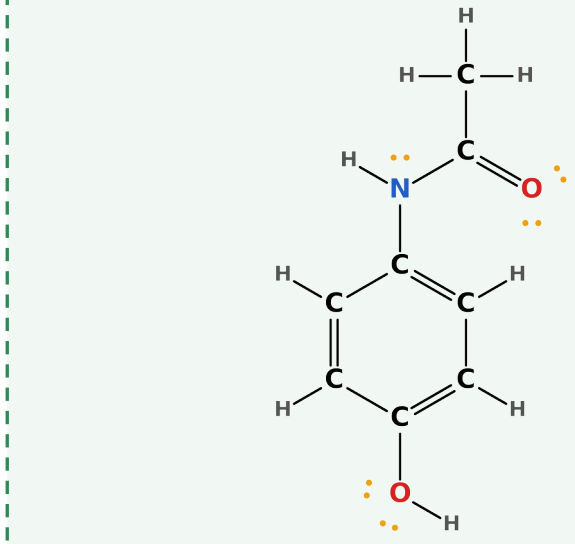
$C_8H_9NO_2$

ATOM COUNT

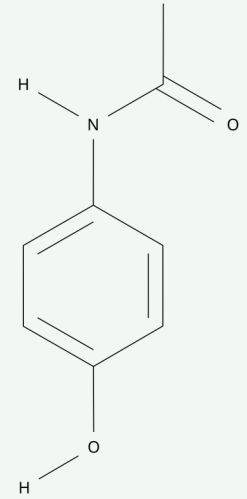
C 8 | H 9 | N 1 | O 2

20 atoms

2 LEWIS STRUCTURE



4 BOND-LINE (KETCHER)



3 CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)NH\text{–}C_6H_4\text{–}OH$$

Ring substituents are para (1,4)

- ✓ atom count verified
- ✓ valence checked
- ✓ bond order checked
- ✓ same connectivity confirmed

ChE connection: Pharmaceutical synthesis and purification

CHEM 13 · MOLECULE CARD

09

Aspirin (acetylsalicylic acid)

IUPAC: 2-acetyloxybenzoic acid

1 MOLECULAR FORMULA

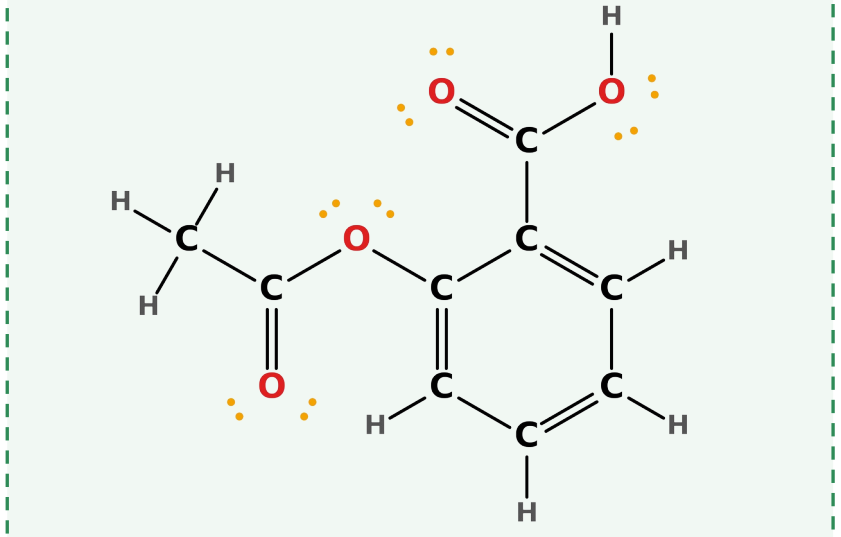
$C_9H_8O_4$

ATOM COUNT

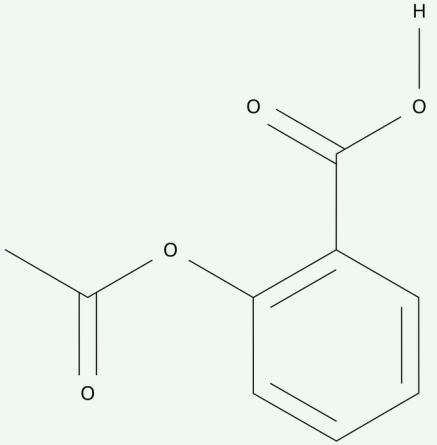
C 9 | H 8 | O 4

21 atoms

2 LEWIS STRUCTURE



4 BOND-LINE (KETCHER)



3 CONDENSED STRUCTURAL FORMULA

$$CH_3C(=O)O\text{–}C_6H_4\text{–}C(=O)OH$$

Ring substituents are ortho (1,2)

- ✓ atom count verified
- ✓ valence checked
- ✓ bond order checked
- ✓ same connectivity confirmed

ChE connection: Pharmaceutical reaction and crystallization

CHEM 13 · MOLECULE CARD

10

Caffeine

IUPAC: 1,3,7-trimethylpurine-2,6-dione

1 MOLECULAR FORMULA

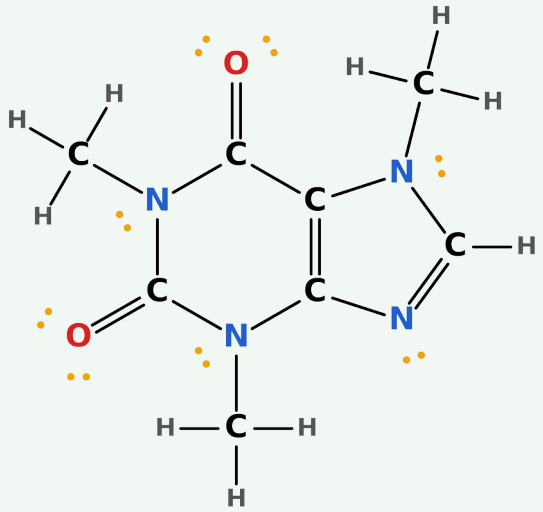
$C_8H_{10}N_4O_2$

ATOM COUNT

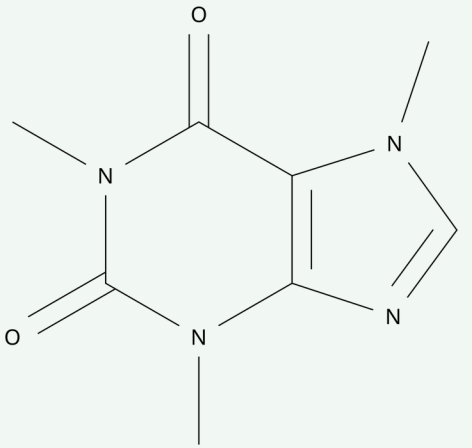
C 8 | H 10 | N 4 | O 2

24 atoms

2 LEWIS STRUCTURE



4 BOND-LINE (KETCHER)



3 CONDENSED STRUCTURAL FORMULA

$CH_3N1C{=}NC2{=}C1C(=O)N(CH_3)C(=O)N2CH_3$

Fused rings, so written as a connectivity string. N–CH_3 at N1, N3, N7; C=O at C2 and C6

- ✓ atom count verified
- ✓ valence checked
- ✓ bond order checked
- ✓ same connectivity confirmed

ChE connection: Extraction, separations, food/pharma processing